Branch Sums

AlgoExpert · Easy · Binary Trees

The problem

- List the root-to-leaf path sums, ordered left to right.

Intuition

- Carry the running sum down each path; at a leaf, that sum is one branch sum.

Approach

- 1. Depth-first from the root with sum zero
- 2. Add the node's value
- 3. At a leaf, record the sum
- 4. Recurse left then right to keep left-to-right order

Java solution

```java
public class BranchSums {
    public static List<Integer> branchSums(BinaryTree root) {
        List<Integer> sums = new ArrayList<>();
        dfs(root, 0, sums);
        return sums;
    }

    private static void dfs(BinaryTree node, int running, List<Integer> sums) {
        if (node == null) return;
        int s = running + node.value;
        if (node.left == null && node.right == null) { sums.add(s); return; }
        dfs(node.left, s, sums);
        dfs(node.right, s, sums);
    }
}
```

Complexity

- Time: O(n)
- Space: O(n)

Recap

- Carry the running sum down each path; at a leaf, that sum is one branch sum.
- Complexity: O(n)